

Arth Banka

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Education

CARNEGIE MELLON UNIVERSITY

Master of Science: Computational Biology | GPA: 4.17 / 4.00

Pittsburgh, PA | Aug 2024 – Dec 2025

Relevant Coursework: Algorithms and Advanced Data Structures, Generative AI, Programming, Statistics, M.S. Research

INDIAN INSTITUTE OF TECHNOLOGY KANPUR

B. Tech in Biological Sciences and Bioengineering | GPA: 8.8/10.0 (Department Rank: 1)

Kanpur, India | Aug 2020 – May 2024

Relevant Coursework: Computational Biology and Bioinformatics, Introduction to Machine Learning, Big-Data Visual Analytics

Skills

Programming: C/C++, Python, R, Shell Scripting, Golang, MATLAB

Bioinformatics: FastQC, STAR, Kraken2, Bedtools, UCSC Tools, IGV, BLAST, DESeq2, MEME suite

Work-flow and Misc. Tools: Linux, Git, Docker, AWS, TensorFlow, PyTorch, Snakemake, MultiQC

Work Experience

STRUCTURAL BIOINFORMATICS INTERN

Carnegie Mellon University, PA | May 2025 – Aug 2025

- Built a large-scale dataset (100k entries) of protein-metal binding residues and designed scalable pipelines for high-throughput sequence-structure analysis using UNIX/Linux environments.
- Implemented a sequence modeling framework integrating pre-trained embeddings with state-space models, demonstrating 94% accurate classification of biologically relevant protein-metal interactions.
- Adapted workflow modules to integrate into production-scale NGS data analysis environments for workflow pipelines.

COMPUTATIONAL RESEARCH INTERN

Dalhousie University, Halifax, Canada | May 2023 – Aug 2023

- Developed randomized algorithms for phylogenetic tree inference, reducing runtime by 66% and enabling large-scale evolutionary analyses relevant to genomic epidemiology, antibiotic resistance and lateral gene transfer.
- Optimized the Phylogenetic Topographer algorithm by implementing a **graph-based data structure** for faster searching and visualization of tree space graphs and presented research to departmental audience.

Project Experience

SUPER ENHANCER & RNA BINDING PROTEIN INTERACTION ANALYSIS

Carnegie Mellon University | Aug 2024 – Aug 2025

- Processed and analyzed large-scale NGS datasets (eCLIP, ChIP-seq, RNA-seq, ATAC-seq, and conservation tracks from ENCODE and NCBI) to interrogate enhancer-RBP interactions across 700 genomic loci revealing a 30% concordance between ChIP and eCLIP peaks in top super enhancers.
- Built a reproducible end-to-end bioinformatics pipeline (QC, alignment, peak calling, annotation, and visualization) using Bedtools, bigWig utilities, and UCSC genome browser to support scalable report generation for downstream analysis.

SKETCH-BASED ANALYSIS OF scRNA-SEQ DATA

Carnegie Mellon University | Dec 2024 – Feb 2025

- Engineered and benchmarked six sketching algorithms (uniform, geometric, k-means, kernel herding, hashing, PCA-guided) on 68k PBMC scRNA-seq dataset to evaluate rare cell capture and transcriptional diversity retention.
- Engineered a variance-aware PCA-guided sketching method reducing data volume by 90% while preserving all biological clusters and recovering 172 of 193 rare cells, enabling scalable transcriptomic analysis on standard hardware.

GENOMICS AND EPIGENETICS OF THE BRAIN

Carnegie Mellon University | Aug 2024 – Dec 2024

- Analyzed single-cell NGS datasets with Python (clustering, dimensionality reduction) and R (epigenetic state analysis, TF binding) and conducted a statistical analysis of neural epigenetic datasets.
- Computationally identified molecular and functional difference between neurons, astrocytes, oligodendrocytes, and microglial cells using CNN's and SVM's with classification accuracy of 91%.

Leadership

Led 30-member team to 1st in aerial systems competitions and co-led a CMU BioHackathon project on **GPU-accelerated multi-omics modeling**, awarded **most innovative & best teamwork** and published in **BioHackRxiv (2025)**.